

Artwork Damage Restoration Assistant using Retrieval-Augmented Generation (RAG)

Problem Statement

Artwork restoration and conservation require specialized knowledge derived from museum guidelines, conservation manuals, research papers, and historical restoration records. Accessing and interpreting this information can be difficult for students, researchers, and conservation professionals due to the large volume of scattered documentation.

Traditional chatbots and language models may generate generic or inaccurate recommendations because they do not have direct access to domain-specific conservation knowledge. This can lead to unreliable guidance when dealing with artwork damage such as cracks, paint loss, discoloration, mold growth, tears, and environmental deterioration.

The objective of this project is to develop a Retrieval-Augmented Generation (RAG) system that retrieves relevant information from a curated knowledge base of conservation resources and uses that information to generate accurate, context-aware responses. The system aims to support artwork conservation by providing reliable restoration guidance, preservation recommendations, and explanations grounded in trusted documentation.